Targeting Higher Credit Card Payments

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Anchoring Heuristic

Anchoring & adjustment heuristic one of Tversky & Kahneman (1974)'s foundational biases

Key idea:

- Anchors: Salient numbers that can bias consumer choices.
- Consumers choose at / near numerical amount initially presented, & adjust away too little.

"One of the most robust phenomena studied across the social behavioral sciences."
- Meta-analysis by Schley and Weingarten (2025)

Anchoring Influential in Credit Card Literature

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- Support for anchoring from studies of credit card administrative data (Keys & Wang, 2019; Medina, & Negrin, 2022; Vihriälä 2025)

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Key implication is that without anchoring, consumers would pay more on their credit cards.

This Paper: Do Online Experiment Findings Extrapole To The Field?

Field experiment on 6,714 credit cardholders with a U.S. FinTech

- RCT varying payment options in their app
  - Test shrouding minimum payment amount
  - Test adding 50% payment amount option

- Observe outcomes for six cycles

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Key Findings: Results of hypothetical online experiments do not translate into the field

- No increase in payments from shrouding the minimum payment
- Adding 50% payment option does not increase payments
- Minimum payment amount is (relevant) target **not** (irrelevant) anchor

Contributions

- Evidence that a **policy** shrouding the minimum payment is expected to be ineffective
  - (Financial Conduct Authority, 2021; Brookings, 2022; Financial Health Network, 2023)

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- **Household Finance:** Why consumers often only pay at/just above the minimum?
  - *Not* driven by heuristic anchoring to the minimum payment
  - Cardholders frequently lack liquid cash & target paying at least minimum payment amount, which acts as a reference point
  - Bunching in payments at/just above minimum driven by discontinuity in costs

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- **Behavioral Economics/Finance:** Weakness of active choice nudges & anchoring heuristic

- **Empirical methods** in social sciences:
  - Online studies increasingly used across social sciences (Rauf et al., 2025)
  - Magnitudes can differ between online and field experiments (Gandhi et al., 2025)
  - Example where not just magnitudes differ, but key finding

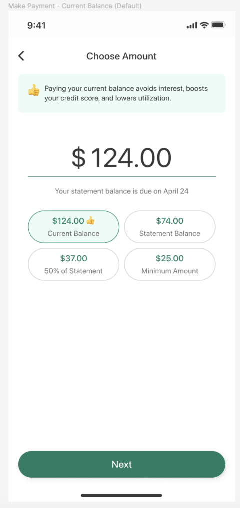
Roadmap For Presentation

1. The Experiment (Design, Data, Methodology)
2. Experimental Results
3. Heterogeneity, Distinguishing Anchoring from Targeting

The Experiment

Field Experiment Design

Control (C)



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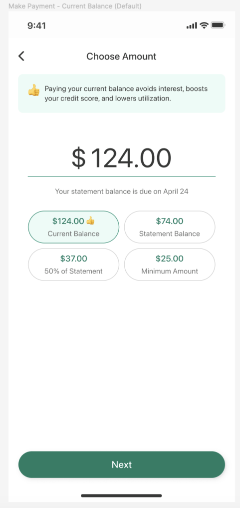
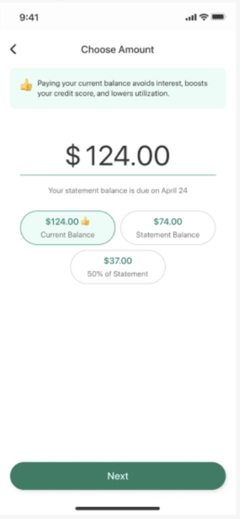


Treatment T1



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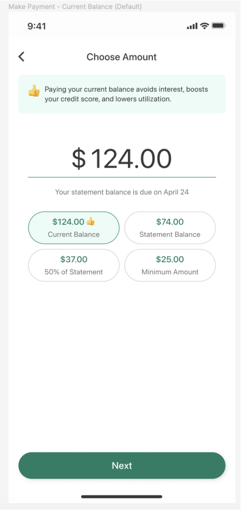

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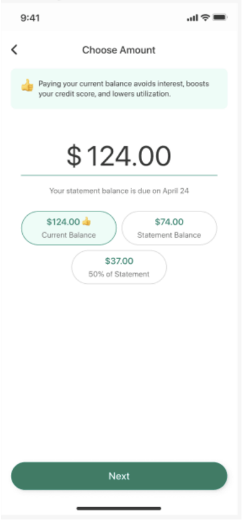
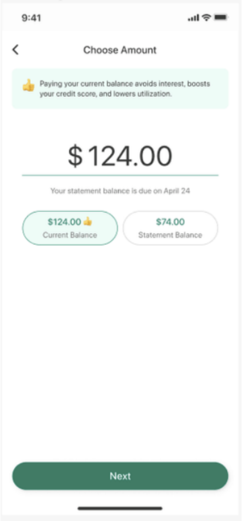


Treatment T2



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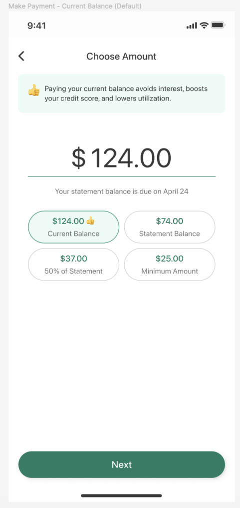

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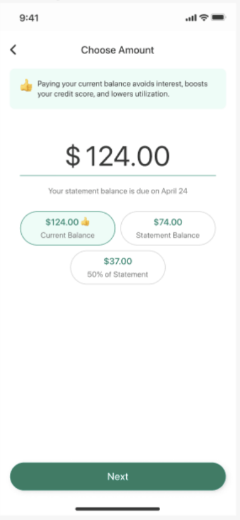
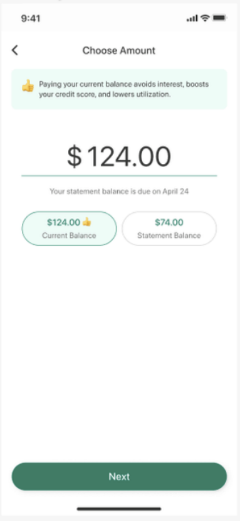
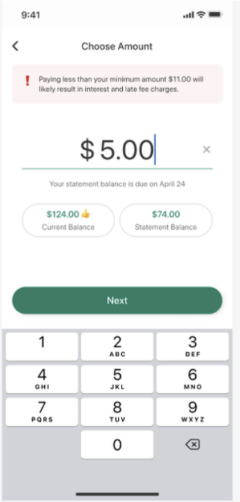


Treatment T2



Prompt



Field Experiment Design

- **Control (C)**, 4,542 consumers:
  minimum payment, 50% statement balance, statement balance, current balance, text box
- **Treatment 1 (T1)**, 1,096 consumers:
  ~~minimum payment,~~ 50% statement balance, statement balance, current balance, text box
- **Treatment 2 (T2)**, 1,076 consumers:
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In C, T1, & T2, if attempt to pay $<$ minimum, a **prompt** appears with minimum payment

T1 (vs C):

Effect of *shrouding* minimum payment option

T2 (vs C):

Effect of *shrouding* minimum payment and 50% statement balance options

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T1 (vs T2):

Effect of *adding* 50% statement balance option (holding constant shrouding of minimum)

Data

Data Provider

U.S. FinTech lender providing credit cards:

- Means: \$290 credit limits, \$200 statement balance, 560 VantageScore.

Below prime ($<$ 660 VantageScore) consumers:

- In Keys & Wang (2019), group most likely to pay at/near minimum payment, similar anchoring estimates to prime consumers

- Largest potential gains from paying more, given highest interest rates (Drechsler et al., 2025; Consumer Financial Protection Bureau, 2025)

Datasets

Linked lender's administrative credit card data with (1) Clickstream data on payment choices, (2) Financial transaction data on liquid cash (Plaid), (3) Equifax credit reporting data

Empirical Methodology

Today, focus on regression results from dynamic specification:

$$Y_{i,t} = \sum_{\tau=0}^{6} \left[\delta_{\tau}^{T1(vsC)} \left(D_{\tau} \times T1_i \right) + \delta_{\tau}^{T2(vsC)} \left(D_{\tau} \times T2_i \right) \right] + X_i' \beta + \mu_t + \varepsilon_{i,t} \qquad (1)$$

X_i vector of pre-experiment controls, μ_t statement cycle fixed effects

Standard errors clustered at consumer-level

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X_i vector of pre-experiment controls, μ_t statement cycle fixed effects

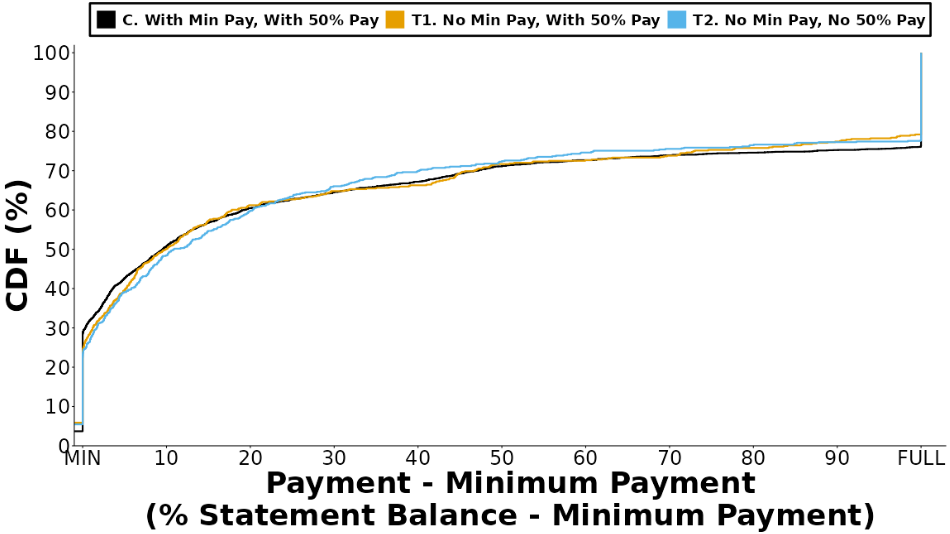
Standard errors clustered at consumer-level

Also, show pooled specification to increase power:

$$Y_{i,t} = \delta^{T1(vsC)}\, T1_i + \delta^{T2(vsC)}\, T2_i + X_i' \beta + \mu_t + \varepsilon_{i,t} \tag{2}$$

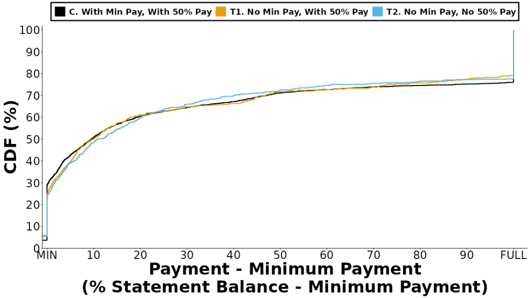
Experimental Results

Payment Distribution Shows Little Overall Change

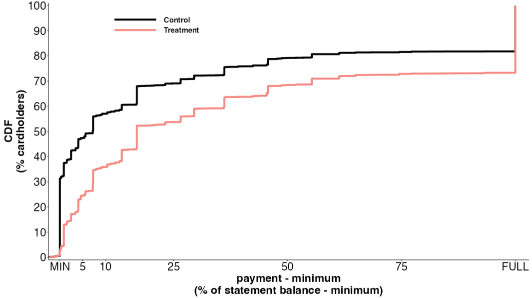


Field Experiment Results In Sharp Contrast To Online Experiments

Field Experiment (This Paper)



Online Experiment Example (Guttman-Kenney, 2024)



$\uparrow$ 12.4*** p.p. (s.e. 0.8)

($N = 7{,}938$, mean 28.1%)

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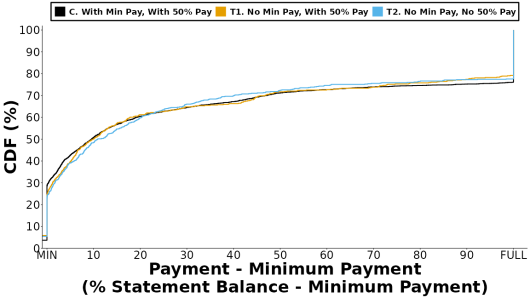

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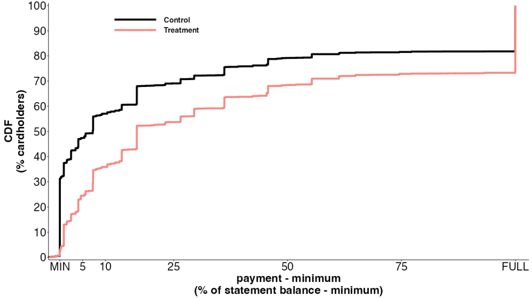

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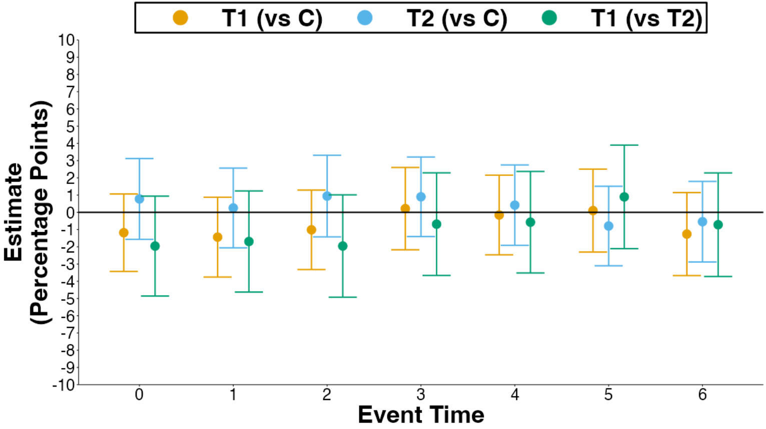
Challenging for online experiments to recreate the frictions, intertemporal dynamics, & high-stakes incentives that consumers experience managing their finances.

Effects On Primary Outcome: Payments (% Statement Balance)



δ_6^k (percentage points, C mean: 27.7%):

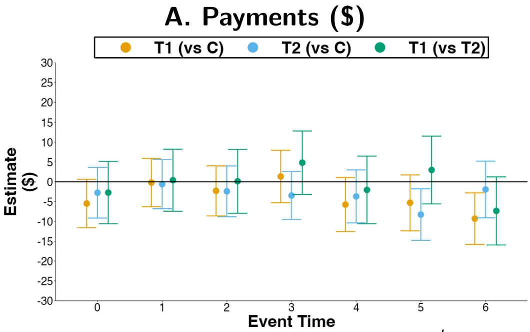
T1 (vs C) -1.26 (s.e. 1.23) **T2 (vs C)** -0.54 (s.e. 1.19) **T1 (vs T2)** -0.72 (s.e. 1.53)

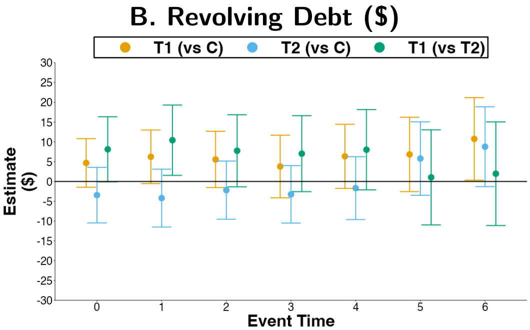
Effects on Payments (% Statement Balance) Pooled Across Cycles 0 to 6

	Pooled (δ^k)			
Comparison	(1)	(2)	(3)	(4)
T1 (vs C)	0.49	0.55	-0.67	-2.39
	(1.06)	(1.06)	(0.81)	(1.23)
T2 (vs C)	0.35	0.37	0.29	1.04
	(1.05)	(1.05)	(0.80)	(1.16)
T1 (vs T2)	0.14	0.18	-0.96	-3.44*
	(1.35)	(1.35)	(1.02)	(1.51)
Time F.E.		X	X	X
Consumer Controls			X	
Consumer F.E.				X

(percentage points, C mean: 34.6%)

Robust To Measuring Payments ($), Revolving Debt ($)



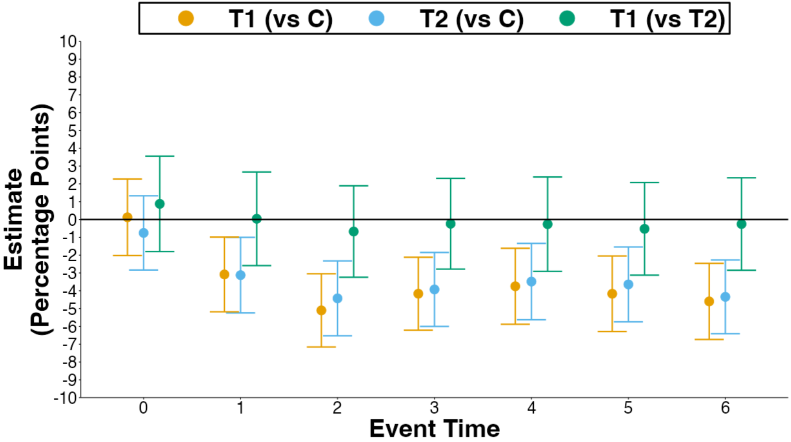


δ_6^k (percentage points):

T1 (vs C) $-\$9.41^*$ (s.e. \$3.32)
T2 (vs C) $-\$1.95$ (s.e. \$3.65)
T1 (vs T2) $-\$7.36$ (s.e. \$4.38)
(C mean: \$58.4)

T1 (vs C): $\$10.76^*$ (s.e. \$5.32)
T2 (vs C): \$8.79 (s.e. \$5.14)
T1 (vs T2): \$1.97 (s.e. \$6.68)
(C mean: \$230.9)

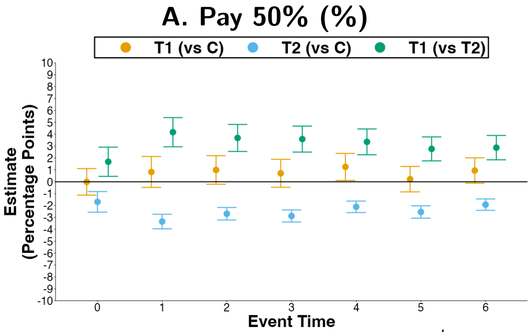
Effects On Payment Distribution: Paying Minimum

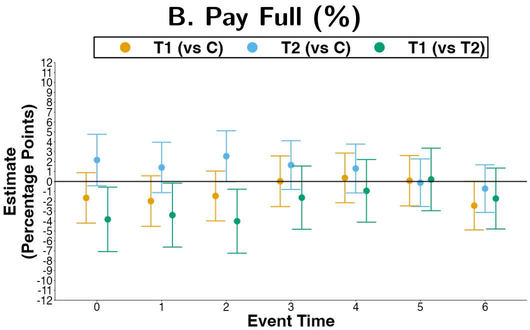


δ_6^k (percentage points, C mean: 15.6%):

T1 (vs C) -4.60*** (s.e. 1.09) **T2 (vs C)** -4.34*** (s.e. 1.06) **T1 (vs T2)** -0.26 (s.e. 1.32)

Effects On Paying 50% and Full Balance

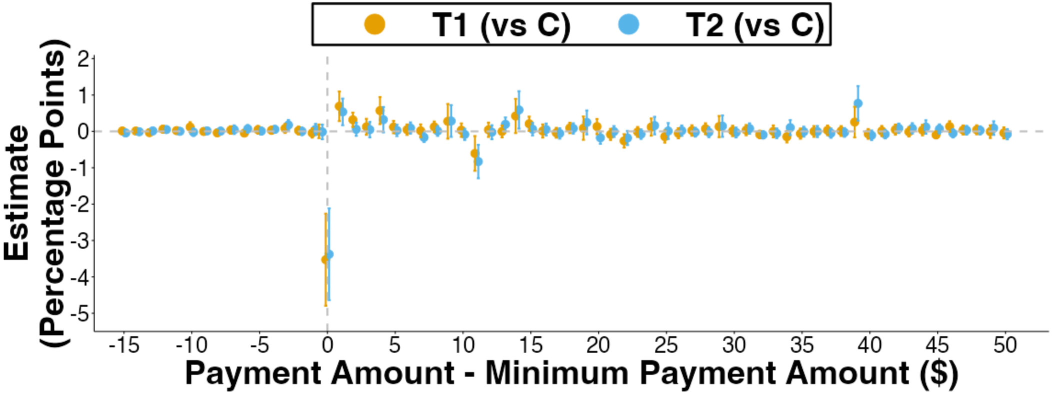




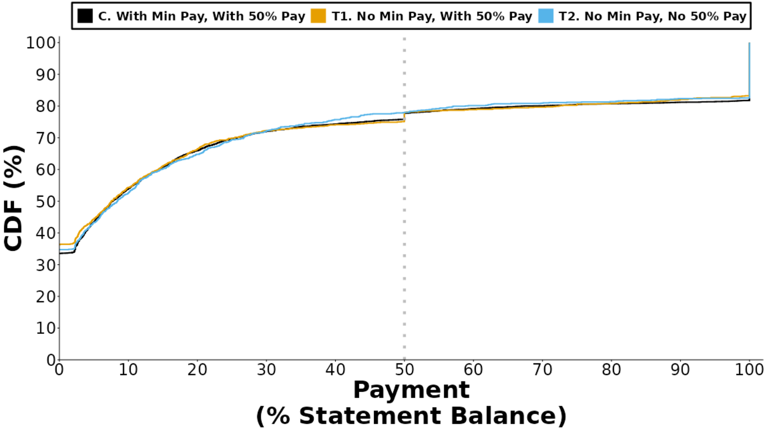
δ_6^k (percentage points):

Pay 50%	Pay Full
T1 (vs C) 0.94 (s.e. 0.54)	**T1 (vs C)** -2.44 (s.e. 1.25)
T2 (vs C) -1.93^{***} (s.e. 0.24)	**T2 (vs C)** 0.72 (s.e. 1.23)
T1 (vs T2) 2.87^{***} (s.e. 0.52)	**T1 (vs T2)** -1.72 (s.e. 1.57)
(C mean: 2.7%)	(C mean: 22.8%)

Effects On Payments Proximate To Minimum Payment Amount



Effects On Payment Distribution



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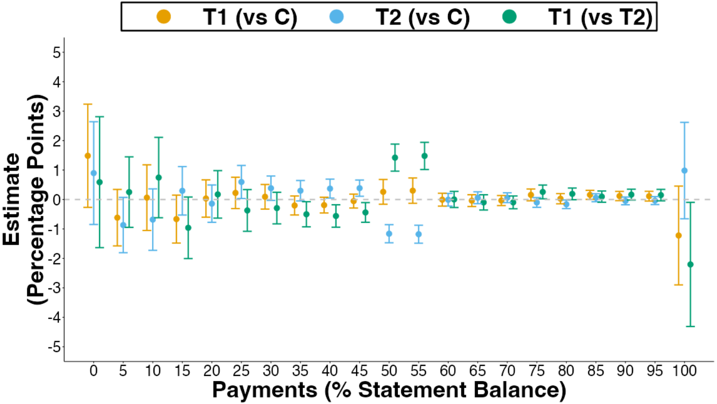
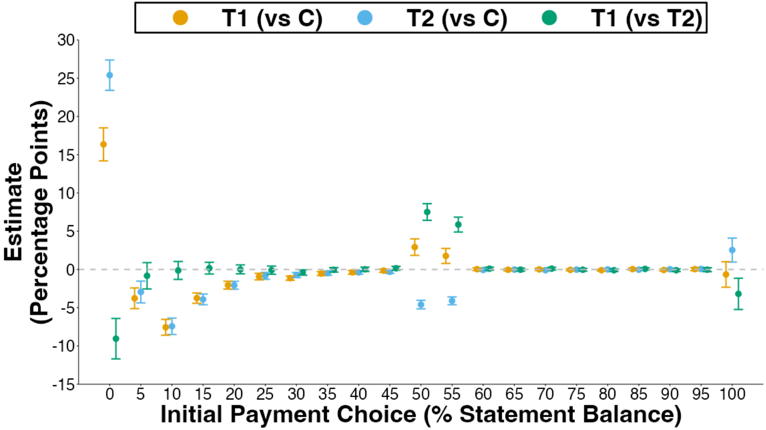


Adding 50% option:

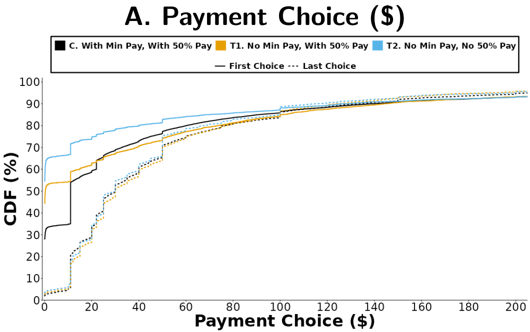
$$\downarrow Pr(Payment \in \{(30\%, 45\%], 100\%\})$$

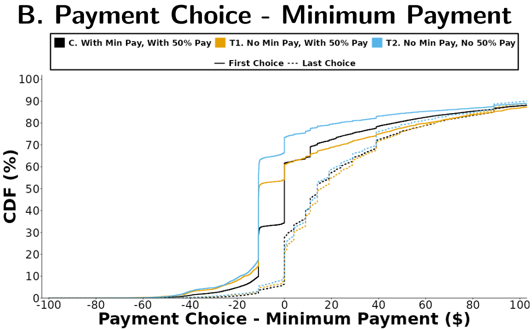
$$\uparrow Pr(Payment \in \{(45\%, 55\%]\})$$

Effects On Initial Payment Choice



Initial vs. Last Payment Choices Within Payment Cycle





Pr (Initial Payment Choice < Minimum Payment):

T1 (vs C) 19.44*** (s.e. 1.16) **T2 (vs C)** 30.50*** (s.e. 1.02)

T1 (vs T2) 11.06*** (s.e. 1.38)

(percentage points, C mean: 33.7%)

Heterogeneity

Test Three Sources of Heterogeneity

1. Near Minimum Payers
2. Credit Card Utilization
3. Liquid Cash

Heterogeneous groups calculated with pre-experiment data.

Follow Keys & Wang (2019) Defining Near Minimum Payers

Anchoring predicts largest results for unconstrained consumers "near minimum payers":

$$\textit{payments} \in \{(\textit{Minimum}, \textit{Minimum} + \$50] \setminus \textit{Full}\}$$

18% my data, 20% in Keys & Wang (2019)

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Sample	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Consumers
Near Minimum	2.89	-0.56	3.45	1,226
	(1.73)	(1.54)	(2.07)	
Not Near Minimum	-1.44	0.28	-1.72	5,488
	(0.92)	(0.92)	(1.17)	
Time F.E.	X	X	X	
Consumer Controls	X	X	X	

Heterogeneous Effects By Pre-Experiment Credit Card Utilization

Portfolio utilization: 25% (mean) & 36% (median). 16% consumers with 90+.

Sample	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Consumers
Q1 Card Utilization (<60%)	-1.89	3.70*	-5.59**	2,238
	(1.65)	(1.60)	(2.08)	
Q2 Card Utilization (60-91%)	1.57	-1.19	2.76	2,238
	(1.40)	(1.31)	(1.72)	
Q3 Card Utilization (91%+)	-1.45	-1.30	-0.15	2,238
	(1.15)	(1.17)	(1.44)	
Q1 Portfolio Utilization (<12%)	-1.68	0.37	-2.05	2,238
	(1.43)	(1.47)	(1.86)	
Q2 Portfolio Utilization (12-45%)	-2.31	0.94	-3.25	2,238
	(1.43)	(1.40)	(1.83)	
Q3 Portfolio Utilization (45%+)	1.66	-0.96	2.62	2,238
	(1.45)	(1.36)	(1.78)	
Time F.E.	X	X	X	
Consumer Controls	X	X	X	

Heterogeneous Effects By Pre-Trial Liquid Cash Balances

Sample	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Consumers
Liquid Cash <$1	1.32	1.01	0.31	1,025
	(2.28)	(2.22)	(2.97)	
Liquid Cash $1+	-1.36	0.20	-1.56	5,689
	(0.88)	(0.86)	(1.10)	
Liquid Cash <$501	-0.46	0.70	-1.16	3,796
	(1.06)	(1.06)	(1.34)	
Liquid Cash $501+	-0.80	-0.45	-0.35	2,918
	(1.3)	(1.23)	(1.60)	
Liquid Cash <$1,001	-1.11	0.42	-1.53	4,726
	(0.95)	(0.95)	(1.21)	
Liquid Cash $1,001+	0.25	-0.22	0.47	1,988
	(1.58)	(1.50)	(1.96)	
Minimum Liquid Cash <$1	-1.47	0.68	-2.15	4,230
	(1.02)	(1.03)	(1.31)	
Minimum Liquid Cash $1+	-0.25	-0.15	-0.09	2,484
	(1.37)	(1.33)	(1.70)	
Minimum Liquid Cash <$501	-0.88	0.61	-1.49	6,252
	(0.83)	(0.83)	(1.06)	
Minimum Liquid Cash $501+	1.50	0.43	1.07	462
	(3.33)	(3.34)	(4.07)	
Minimum Liquid Cash <$1,001	-0.95	0.21	-1.16	6,422
	(0.82)	(0.82)	(1.04)	
Minimum Liquid Cash $1,001+	1.77	5.87	-4.10	292
	(4.46)	(4.55)	(5.77)	
Time F.E.	X	X	X	
Consumer Controls	X	X	X	

93% consumers with *minimum* liquid cash balance <$501 in 90 days pre-experiment

Implications for macroeconomic models

- Heterogeneous agent models include hand-to-mouth consumers with zero liquid wealth **experiencing binding borrowing constraints** (e.g., Kaplan & Violante, 2022)
- However, Lee and Maxted (2025) show that **binding borrowing constraints is inconsistent with SCF data**:
  - Consumers often borrow intermediate levels of credit card debt, *not* borrowing constrained!
- My results show Lee and Maxted (2025)'s results are not due to under-reporting of credit card debt in SCF (Zinman, 2009)
- My empirical support for Lee and Maxted (2025)'s theory has broad implications, as it affects how households respond to fiscal and monetary policy

Distinguishing Anchoring from Targeting

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Anchors: Irrelevant numbers influence choices e.g., thinking of your (randomly-assigned) last two digits of SSN influences choices (Tversky & Kahneman, 1974; Ariely et al., 2003)

Targets (reference-dependent preferences): Consumers exert effort to meet/exceed relevant numbers e.g., marathon time goals (Heath, Larrick, & Wu, 1999; Allen et al., 2017)

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Distinguish using **"VPA" Value Proximity Analysis** (Bartels, Herzog, & Sussman, 2024)

$$Y_i = \frac{1}{N_i} \sum_{j \in S_i} 1(V_j \geq F)$$

$$Y_i = \alpha + \beta^{T1}\, T1_i + \beta^{T2}\, T2_i + \gamma Proximity_i + \theta^{T1}\, (Proximity_i \times T1) + \theta^{T2}\, (Proximity_i \times T2) + \varepsilon_i$$

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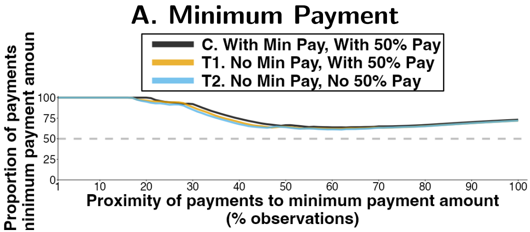
VPA Predicts:

Anchoring: $\alpha \approx 50\%$, $\gamma >$ if low anchor, $\gamma < 0$ if high anchor, $\beta^{T1}, \beta^{T2}, \theta^{T1}, \theta^{T2} \neq 0$

Targeting: $\alpha \approx 100\%$, $\gamma < 0$, $\beta^{T1}, \beta^{T2}, \theta^{T1}, \theta^{T2} = 0$

Value Proximity Analysis (VPA): Payments Proximate To Focal Amounts



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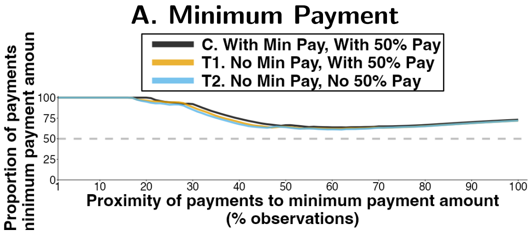

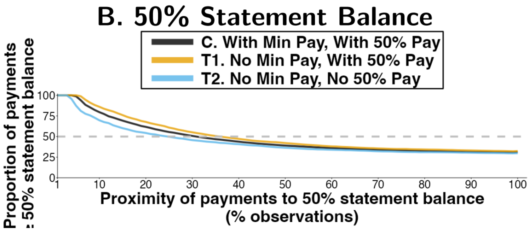

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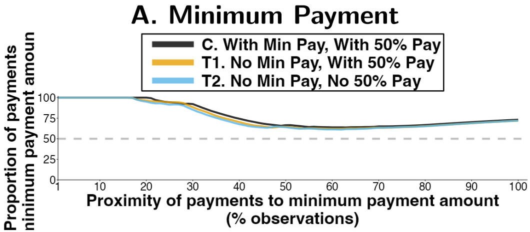

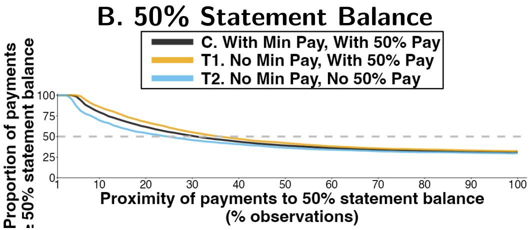

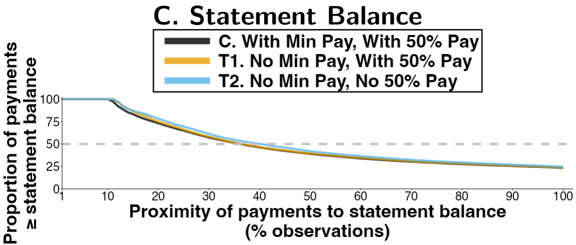

VPA Results Inconsistent With Anchoring

	Minimum Payment (1)	50% Statement Balance (2)	Statement Balance (3)
Intercept	98.510***	76.831***	89.091***
	(1.411)	(2.718)	(2.308)
T1	−1.403	5.290	1.380
	(2.121)	(3.645)	(3.244)
T2	−2.840	−7.568	3.377
	(2.212)	(3.986)	(3.052)
Proximity	−0.406***	−0.584***	−0.798***
	(0.028)	(0.042)	(0.037)
Proximity x T1	−0.005	−0.044	−0.012
	(0.041)	(0.057)	(0.052)
Proximity x T2	0.008	0.078	−0.016
	(0.042)	(0.062)	(0.050)
Observations	300	300	300
Adjusted R^2	0.647	0.773	0.873

Evidence Against Credit Card Anchoring

1. Shrouding minimum payment does not increase payments
2. Heterogeneity analysis shows no increase in payments for near minimum payers
3. Initial payment choices are lower when minimum payment is shrouded
4. Value Proximity Analysis results inconsistent with anchoring

(+ shrouding autopay minimum payment option also ineffective, Guttman-Kenney et al., 2025)

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Instead, makes sense for consumers to target minimum payment, given frequently low liquid cash balances & discontinuity in economic payoffs generating observed bunching in payments.

"An issue that has received relatively little attention in the literature is that kinks and notches may represent reference points" (Kleven, 2016)

Reconciling Results With Credit Card Literature

- Well-established that credit card behavior is hard-to-change.
- Lack of sophistication alone cannot explain this phenomenon, as information to debias consumers does not work in this domain.
- Understand why other suggested payment amounts (e.g., 50%, CARD Act scenarios) ineffective.
  - Intermediate payment values struggle to compete with consumers targeting paying at least the minimum and the full amount, where strong economic incentives exist.
- Understand why lenders set low minimum paymetns and why increases to minimum payments can be effective to reduce debt.

(Beshears et al., 2018; Agarwal et al., 2015; Keys & Wang, 2016, 2019; Seira et al., 2017; Medina, 2021; Adams et al., 2022; Medina & Negrin, 2022; Katz et al., 2024; Castellanos et al., 2025; Guttman-Kenney et al., 2025; Batista et al., 2025; Han & Yin, 2025; Allen et al., 2026)

Conclusions

Key Findings

- Shrouding the minimum payment does *not* increase in credit card payments
- Adding 50% payment option also does *not* increase credit card payments
- Consumers frequently have low liquid cash balances (but without borrowing constraint)

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Key Implications

- **Credit Cards:** Below prime credit cardholders frequently have low liquid cash balances.
  - Minimum payment amount is (relevant) target **not** (irrelevant) anchor.
- **Behavioral Economics:** Consumers "hack" active choice nudge.
- **Macroeconomic Models:** Hand-to-mouth consumers often not borrowing constrained.
- **Survey Methods:** Use online experiments with caution!

Thank you!

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Pre-Experiment Controls

Linear controls:

- Age
- Credit Score
- Card Tenure
- Pre-experiment values of all of the primary and secondary outcomes
- Number of open credit cards held in their portfolio
- Total credit card portfolio balances
- Total credit card portfolio credit limits

Fixed effects:

- Interest rate
- Credit limit
- Autopay type

Primary and Secondary Outcomes

Primary Outcome: 1. Payments (%) $= \sum$ **Payments / Statement Balance)**

Secondary Outcomes:

2. Payments ($)
3. Statement Balance ($)
4. Spending ($)
5. Spending (%)
6. Revolving Debt ($)
7. Credit Limit ($)
8. Miss Pay (%)
9. Pay Min (%)
10. Pay 50% (%)
11. Pay Full (%)
12. Pay Current Balance (%)

Pre-Experiment Summary Statistics

Outcome	C Mean (1)	T1 Mean (2)	T2 Mean (3)	T1 - C (4)	T2 - C (5)
Age	45.51	45.68	45.84	0.17	0.33
				(0.38)	(0.39)
				[0.652]	[0.390]
Credit Score	561.70	561.06	562.10	-0.64	0.40
				(2.06)	(2.10)
				[0.757]	[0.847]
APR (%)	15.71	15.72	15.72	0	0
				(0.01)	(0.01)
				[0.826]	[0.85]
Tenure (days)	265.58	257.53	262.24	-8.05	-3.34
				(5.10)	(5.10)
				[0.114]	[0.512]
Autopayer (%)	41.19	43.59	39.69	2.39	-1.50
				(1.68)	(1.65)
				[0.154]	[0.362]
Payments (%)	45.06	48.00	44.34	2.95	-0.72
				(1.44)	(1.39)
				[0.040]	[0.608]
Payments	77.07	85.00	76.99	7.93	-0.08
				(4.21)	(3.79)
				[0.060]	[0.984]
Statement Balance	207.94	206.38	208.75	-1.55	0.82
				(7.57)	(7.41)
				[0.837]	[0.912]
Spending	98.13	99.81	101.3	1.68	3.17
				(4.67)	(4.82)
				[0.719]	[0.510]
Spending (%)	50.11	51.95	50.25	1.84	0.14
				(1.46)	(1.44)
				[0.207]	[0.925]
Revolving Debt	149.09	144.62	149.98	-4.47	0.89
				(7.15)	(6.98)
				[0.532]	[0.898]
Credit Limit	292.09	287.64	294.11	-4.45	2.02
				(8.16)	(8.13)
				[0.585]	[0.803]
Miss Pay (%)	17.00	14.59	15.88	-2.41	-1.12
				(1.21)	(1.24)
				[0.047]	[0.365]
Pay Min (%)	17.48	17.38	16.79	-0.10	-0.69
				(1.29)	(1.26)
				[0.937]	[0.583]
Pay 50% (%)	3.74	3.07	4.20	-0.68	0.45
				(0.60)	(0.67)
				[0.257]	[0.497]
Pay Full (%)	31.73	34.39	30.02	2.66	-1.71
				(1.61)	(1.55)
				[0.098]	[0.270]
Pay Current Balance (%)	9.25	11.15	9.67	1.91	0.42
				(1.05)	(0.99)
				[0.070]	[0.668]

T-Tests (3 Cycles)

Outcome	C Mean (1)	T1 Mean (2)	T2 Mean (3)	T1 - C (4)	T2 - C (5)	T1 - T2 (6)
Payments (%)	32.92	34.4	33.83	1.48	0.92	0.57
				(1.39)	(1.37)	(1.76)
				[0.285]	[0.503]	[0.747]
Payments	59.01	63.18	55.75	4.16	-3.26	7.42
				(4.00)	(3.61)	(4.81)
				[0.297]	[0.367]	[0.123]
Statement Balance	231.01	233.85	225.52	2.84	-5.49	8.33
				(7.93)	(7.65)	(9.89)
				[0.720]	[0.473]	[0.400]
Spending	71.94	73.64	67.50	1.70	-4.43	6.14
				(4.54)	(4.27)	(5.57)
				[0.708]	[0.299]	[0.271]
Spending (%)	36.03	38.04	36.55	2.01	0.52	1.49
				(1.43)	(1.42)	(1.82)
				[0.158]	[0.714]	[0.412]
Revolving Debt	187.21	187.88	182.58	0.67	-4.63	5.30
				(7.70)	(7.48)	(9.65)
				[0.931]	[0.536]	[0.583]
Credit Limit	320.21	319.02	321.35	-1.20	1.14	-2.34
				(8.75)	(8.64)	(11.02)
				[0.891]	[0.895]	[0.832]
Miss Pay (%)	28.82	29.58	29.09	0.76	0.28	0.49
				(1.55)	(1.53)	(1.96)
				[0.622]	[0.857]	[0.803]
Pay Min (%)	16.20	11.63	11.62	-4.57	-4.58	0.01
				(1.12)	(1.11)	(1.38)
				[0]	[0]	[0.995]
Pay 50% (%)	2.72	3.35	0	0.63	-2.72	3.35
				(0.60)	(0.24)	(0.55)
				[0.293]	[0]	[0]
Pay Full (%)	22.76	24.19	24.25	1.42	1.48	-0.06
				(1.45)	(1.44)	(1.84)
				[0.326]	[0.303]	[0.974]
Pay Current Balance (%)	5.90	5.86	6.50	-0.04	0.60	-0.64
				(0.80)	(0.82)	(1.03)
				[0.960]	[0.470]	[0.539]

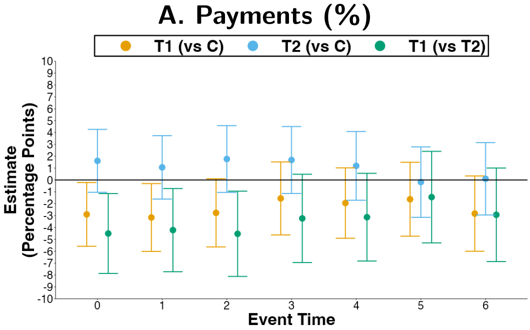
Attrition (3, 6 Cycles)

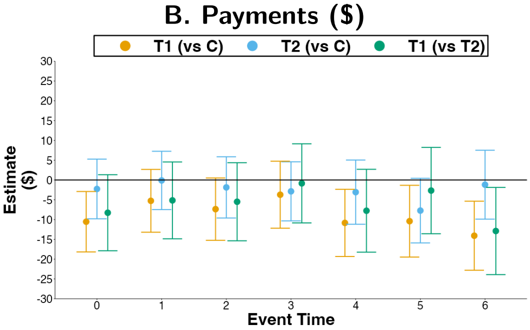
	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)
3	-0.23	-0.12	-0.12
	(0.14)	(0.18)	(0.18)
6	-0.09	0.00	-0.09
	(0.64)	(0.70)	(0.83)
Time F.E.	X	X	X
Consumer Controls	X	X	X

Dynamic Effects on Payments (% Statement Balance) After 3 & 6 Cycles

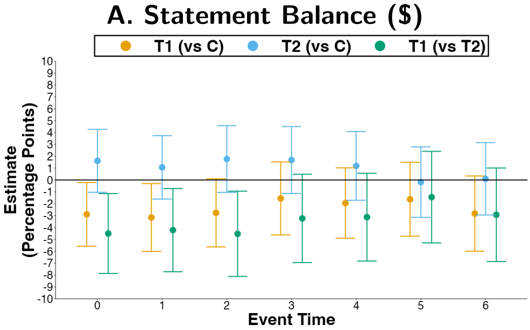
	Dynamic (δ_3^k)				Dynamic (δ_6^k)			
Comparison	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
T1 (vs C)	-0.21	1.47	0.22	-1.55	-7.09***	0.01	-1.26	-2.83
	(1.33)	(1.39)	(1.22)	(1.57)	(1.26)	(1.30)	(1.23)	(1.62)
T2 (vs C)	-0.78	1.00	0.90	1.68	-7.42***	-0.38	-0.54	0.10
	(1.31)	(1.37)	(1.18)	(1.44)	(1.24)	(1.29)	(1.19)	(1.56)
T1 (vs T2)	0.57	0.47	-0.69	-3.23	0.33	0.40	-0.72	-2.93
	(1.75)	(1.75)	(1.52)	(1.90)	(1.64)	(1.64)	(1.53)	(2.01)
Time F.E.		X	X	X		X	X	X
Consumer Controls			X				X	
Consumer F.E.				X				X

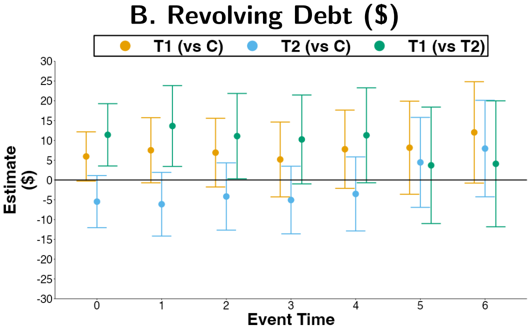
With Consumer F.E.: Payments (%, $)





With Consumer F.E.: Statement Balance ($) & Revolving Debt ($)





Effects on Secondary Outcomes After 3 Cycles

Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)
Payments	1.33	-3.48	4.81
	(3.37)	(3.08)	(4.08)
Statement Balance	5.10	-4.48	9.57*
	(3.65)	(3.38)	(4.48)
Spending	-2.69	-4.57	1.88
	(3.68)	(3.60)	(4.59)
Spending (%)	0.09	0.39	-0.31
	(1.20)	(1.19)	(1.51)
Revolving Debt	3.79	-3.25	7.03
	(4.03)	(3.70)	(4.90)
Credit Limit	0.39	-0.50	0.89
	(2.20)	(2.05)	(2.75)
Miss Pay (%)	1.42	0.45	0.98
	(1.47)	(1.44)	(1.85)
Pay Min (%)	-4.17***	-3.93***	-0.24
	(1.04)	(1.06)	(1.3)
Pay 50% (%)	0.71	-2.88***	3.59***
	(0.60)	(0.26)	(0.56)
Pay Full (%)	0.02	1.66	-1.64
	(1.31)	(1.26)	(1.63)
Pay Current Balance (%)	-0.32	0.60	-0.92
	(0.78)	(0.80)	(1.01)
Time F.E.	X	X	X
Consumer Controls	X	X	X

Dynamic Effects on Secondary Outcomes After 3 Cycles

	T1 (vs C)			T2 (vs C)			T1 (vs T2)		
Outcome	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Payments	-0.07	4.03	-3.72	-7.50*	-3.09	-2.87	7.42	7.12	-0.85
	(3.84)	(4.00)	(4.32)	(3.44)	(3.61)	(3.81)	(4.81)	(4.82)	(5.10)
Statement Balance	-3.34	2.45	4.61	-11.67	-6.00	-5.67	8.33	8.45	10.28*
	(7.84)	(7.89)	(4.45)	(7.55)	(7.63)	(3.79)	(9.89)	(9.87)	(5.19)
Spending	1.07	1.07	-0.42	-5.07	-4.77	-8.11	6.14	5.84	7.69
	(4.35)	(4.53)	(4.80)	(4.06)	(4.27)	(4.92)	(5.57)	(5.58)	(6.14)
Spending (%)	2.46	1.93	-0.01	0.97	0.42	0.26	1.49	1.51	-0.27
	(1.37)	(1.43)	(1.60)	(1.36)	(1.42)	(1.58)	(1.82)	(1.81)	(2.02)
Revolving Debt	-1.39	0.23	5.18	-6.69	-5.17	-5.06	5.30	5.39	10.24
	(7.58)	(7.65)	(4.82)	(7.36)	(7.46)	(4.36)	(9.65)	(9.62)	(5.73)
Credit Limit	-5.61	-2.84	2.18	-3.28	-0.26	-0.84	-2.34	-2.58	3.02
	(8.72)	(8.68)	(2.68)	(8.61)	(8.56)	(2.45)	(11.02)	(10.94)	(3.31)
Miss Pay (%)	2.79	0.84	3.24	2.30	0.38	1.29	0.49	0.46	1.95
	(1.46)	(1.53)	(1.78)	(1.44)	(1.51)	(1.69)	(1.96)	(1.94)	(2.20)
Pay Min (%)	-4.76***	-4.70***	-4.62***	-4.77***	-4.69***	-3.84**	0.01	-0.01	-0.78
	(1.05)	(1.12)	(1.42)	(1.04)	(1.11)	(1.46)	(1.38)	(1.37)	(1.80)
Pay 50% (%)	0.75	0.68	1.36	-2.59***	-2.74***	-3.23***	3.35***	3.43***	4.59***
	(0.56)	(0.60)	(0.8)	(0.12)	(0.25)	(0.71)	(0.55)	(0.56)	(0.95)
Pay Full (%)	0.36	1.33	-1.45	0.42	1.57	3.22*	-0.06	-0.24	-4.66*
	(1.39)	(1.45)	(1.77)	(1.38)	(1.44)	(1.60)	(1.84)	(1.84)	(2.13)
Pay Current Balance (%)	-0.65	-0.08	-1.83	-0.01	0.55	0.18	-0.64	-0.63	-2.01
	(0.74)	(0.80)	(1.23)	(0.77)	(0.83)	(1.19)	(1.03)	(1.04)	(1.55)
Time F.E.		X	X		X	X		X	X
Consumer F.E.			X			X			X

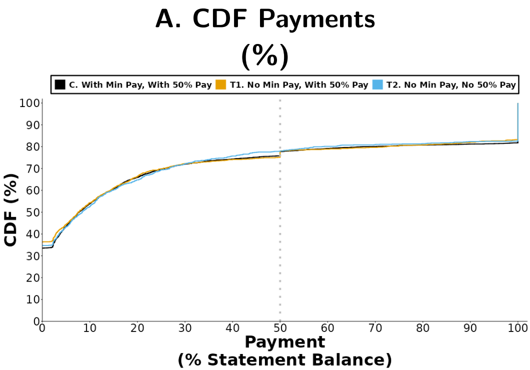
Dynamic Effects on Secondary Outcomes After 6 Cycles

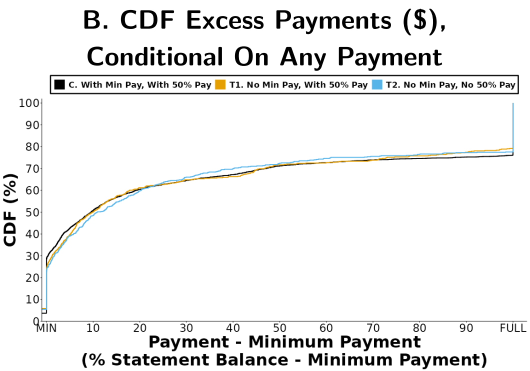
	T1 (vs C)		T2 (vs C)		T1 (vs T2)	
Outcome	(1)	(2)	(3)	(4)	(5)	(6)
Payments	-9.31**	-14.08***	-1.95	-1.19	-7.36	-12.89*
	(3.32)	(4.45)	(3.65)	(4.44)	(4.38)	(5.62)
Statement Balance	6.03	5.54	3.92	3.77	2.11	1.78
	(4.96)	(6.04)	(4.94)	(5.69)	(6.33)	(7.46)
Spending	-6.83	-4.28	-3.27	-6.73	-3.56	2.45
	(4.27)	(5.48)	(3.99)	(5.51)	(5.27)	(7.00)
Spending (%)	-0.41	-0.26	0.71	0.52	-1.12	-0.78
	(1.28)	(1.75)	(1.22)	(1.64)	(1.6)	(2.16)
Revolving Debt	10.76*	12.02	8.79	7.94	1.97	4.09
	(5.32)	(6.53)	(5.14)	(6.22)	(6.68)	(8.12)
Credit Limit	-2.15	-0.14	0.35	0.08	-2.50	-0.22
	(4.33)	(4.97)	(4.17)	(4.74)	(5.47)	(6.22)
Miss Pay (%)	4.39**	5.70***	1.92	2.71	2.47	2.99
	(1.63)	(1.91)	(1.61)	(1.88)	(2.06)	(2.41)
Pay Min (%)	-4.60***	-4.82***	-4.34***	-3.99**	-0.26	-0.84
	(1.09)	(1.56)	(1.06)	(1.50)	(1.32)	(1.93)
Pay 50% (%)	0.94	1.60*	-1.93***	-2.20***	2.87***	3.80***
	(0.54)	(0.80)	(0.24)	(0.71)	(0.52)	(0.95)
Pay Full (%)	-2.44	-3.80*	-0.72	0.66	-1.72	-4.46*
	(1.25)	(1.78)	(1.23)	(1.70)	(1.57)	(2.21)
Pay Current Balance (%)	-0.48	-1.96	-0.33	-0.79	-0.16	-1.18
	(0.66)	(1.17)	(0.66)	(1.17)	(0.84)	(1.49)
Time F.E.	X	X	X	X	X	X
Consumer Controls	X		X		X	
Consumer F.E.		X		X		X

Pooled Effects on Secondary Outcomes

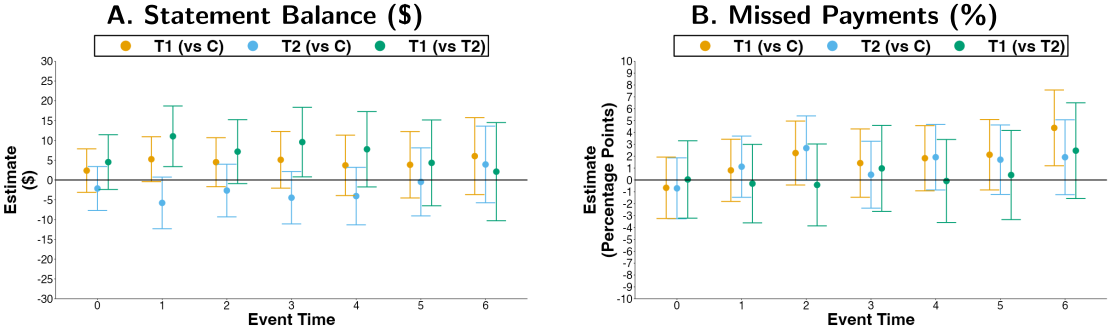
	T1 (vs C)		T2 (vs C)		T1 (vs T2)	
Outcome	(1)	(2)	(3)	(4)	(5)	(6)
Payments	-3.83	-8.85*	-3.31	-2.73	-0.52	-6.12
	(2.19)	(3.43)	(2.13)	(3.12)	(2.74)	(4.15)
Statement Balance	4.39	3.85	-2.29	-3.40	6.68	7.25
	(2.42)	(3.63)	(2.35)	(3.08)	(3.04)	(4.21)
Spending	-2.49	-0.25	-2.32	-5.87	-0.17	5.61
	(2.13)	(3.93)	(2.10)	(4.11)	(2.70)	(5.12)
Spending (%)	-0.38	-0.46	0.72	0.56	-1.10	-1.02
	(0.71)	(1.29)	(0.71)	(1.25)	(0.9)	(1.61)
Revolving Debt	6.29*	7.61	-0.10	-1.81	6.39	9.42*
	(2.78)	(3.95)	(2.72)	(3.64)	(3.50)	(4.77)
Credit Limit	-0.89	0.91	-0.09	-0.41	-0.80	1.32
	(1.72)	(2.26)	(1.64)	(2.11)	(2.17)	(2.80)
Miss Pay (%)	1.72	3.43*	1.29	2.18	0.43	1.26
	(0.90)	(1.34)	(0.90)	(1.30)	(1.15)	(1.66)
Pay Min (%)	-3.53***	-3.94***	-3.38***	-3.29**	-0.15	-0.65
	(0.65)	(1.17)	(0.64)	(1.18)	(0.80)	(1.49)
Pay 50% (%)	0.70*	1.35*	-2.46***	-2.80***	3.16***	4.15***
	(0.29)	(0.62)	(0.15)	(0.67)	(0.28)	(0.81)
Pay Full (%)	-1.00	-2.43	1.20	2.75*	-2.20*	-5.18***
	(0.85)	(1.42)	(0.82)	(1.30)	(1.07)	(1.72)
Pay Current Balance (%)	-0.67	-2.17*	0.26	-0.16	-0.93	-2.01
	(0.40)	(1.01)	(0.42)	(0.97)	(0.52)	(1.26)
Time F.E.	X	X	X	X	X	X
Consumer Controls	X		X		X	
Consumer F.E.		X		X		X

Distribution of Payments, After Six Cycles

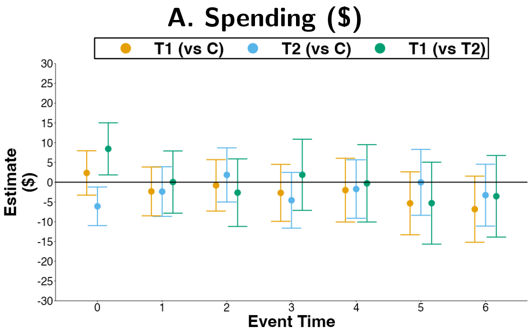




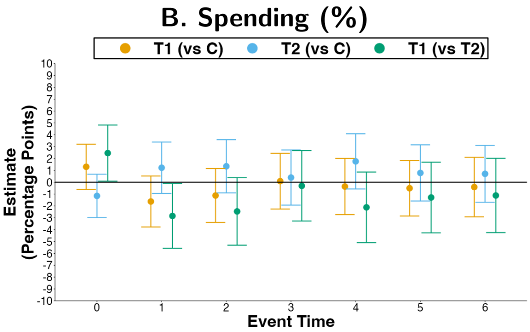
Statement Balance ($), Missed Payments (%)



Spending (%, $)

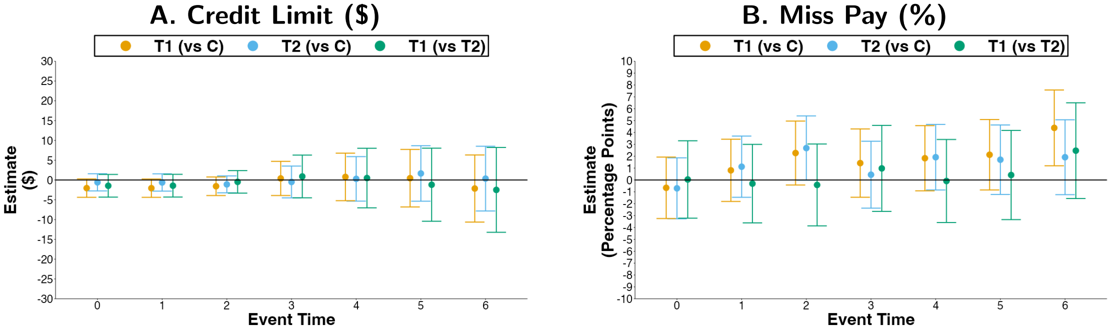


A. Spending ($)
T1 (vs C)
T2 (vs C)
T1 (vs T2)
Estimate
($)
30
25
20
15
10
5
0
-5
-10
-15
-20
-25
-30
0
1
2
3
4
5
6
Event Time

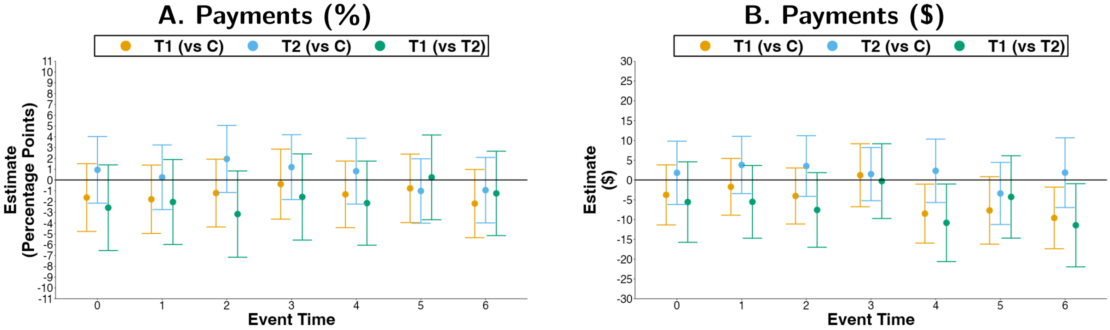


B. Spending (%)
T1 (vs C)
T2 (vs C)
T1 (vs T2)
Estimate
(Percentage Points)
10
9
8
7
6
5
4
3
2
1
0
-1
-2
-3
-4
-5
-6
-7
-8
-9
-10
0
1
2
3
4
5
6
Event Time

Credit Limit ($) and Miss Pay (%)



No Autopay Subsample: Payments (%, $)



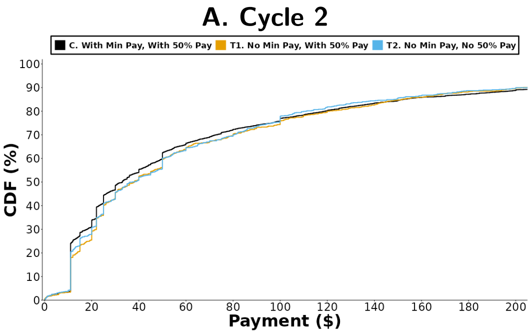
No Autopay Subsample: Dynamic Effects After 6 Cycles

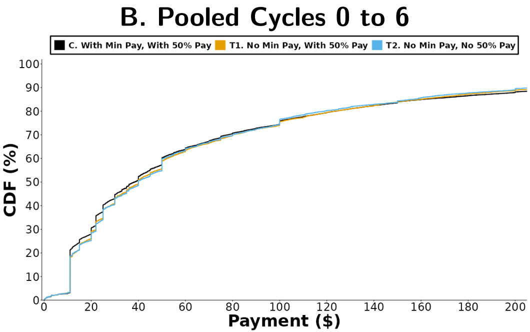
Outcome	T1 (vs C) (1)	(2)	T2 (vs C) (3)	(4)	T1 (vs T2) (5)	(6)
Payments (%)	-2.18	-2.97	-0.94	-0.24	-1.24	-2.73
	(1.61)	(2.08)	(1.55)	(2.02)	(1.99)	(2.58)
Payments	-9.58*	-10.48*	1.85	3.84	-11.44*	-14.31*
	(3.97)	(5.08)	(4.50)	(5.37)	(5.36)	(6.51)
Statement Balance	5.66	3.30	8.07	8.87	-2.41	-5.57
	(5.8)	(6.93)	(5.75)	(6.78)	(7.42)	(8.82)
Spending	-9.12	-7.76	1.15	0.10	-10.27	-7.86
	(5.10)	(6.94)	(5.00)	(6.69)	(6.48)	(8.65)
Spending (%)	-1.12	-0.42	0.03	-0.39	-1.15	-0.03
	(1.73)	(2.37)	(1.58)	(2.15)	(2.11)	(2.89)
Revolving Debt	10.31	9.85	9.13	7.11	1.18	2.74
	(6.09)	(7.46)	(6.04)	(7.43)	(7.76)	(9.47)
Credit Limit	-3.17	-2.24	5.93	6.46	-9.09	-8.70
	(5.2)	(5.78)	(5.20)	(5.89)	(6.73)	(7.50)
Miss Pay (%)	5.03*	5.38*	2.01	2.77	3.02	2.61
	(2.21)	(2.55)	(2.12)	(2.42)	(2.75)	(3.16)
Pay Min (%)	-4.99***	-5.17***	-5.63***	-6.05***	0.64	0.88
	(1.24)	(1.93)	(1.18)	(1.77)	(1.46)	(2.33)
Pay 50% (%)	0.81	2.22	-2.66***	-3.05***	3.47***	5.26***
	(0.81)	(1.15)	(0.38)	(1.08)	(0.78)	(1.41)
Pay Full (%)	-3.84*	-3.81	-1.33	0.43	-2.51	-4.24
	(1.62)	(2.35)	(1.58)	(2.22)	(2.01)	(2.89)
Pay Current Balance (%)	-0.29	-0.96	-0.45	-0.42	0.16	-0.54
	(0.92)	(1.53)	(0.88)	(1.52)	(1.13)	(1.94)
Time F.E.	X	X	X	X	X	X
Consumer Controls	X		X		X	
Consumer F.E.		X		X		X

Effects On Portfolio of Credit Card Statement Balances

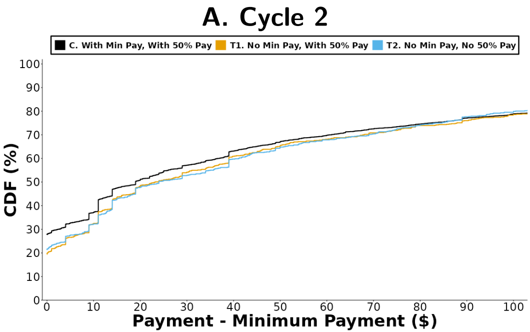
Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)
Portfolio Statement Balances	36.55	6.60	29.94
	(51.85)	(53.41)	(66.98)
Time F.E.	X	X	X
Consumer Controls	X	X	X

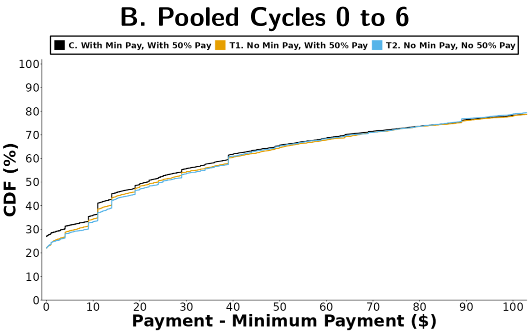
Distribution of Payments ($)



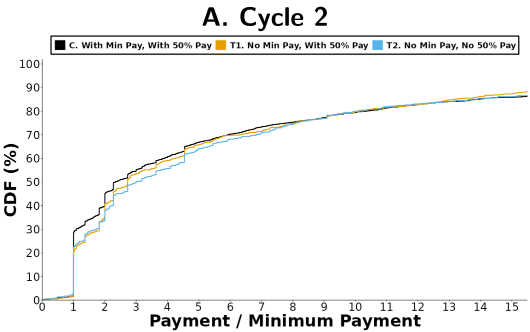


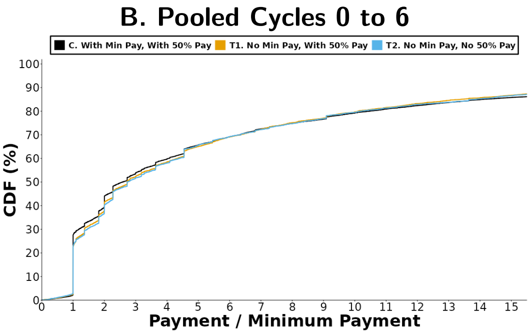
Distribution of Excess Payments ($)





Distribution of Ratio of Payments To Minimum Payment

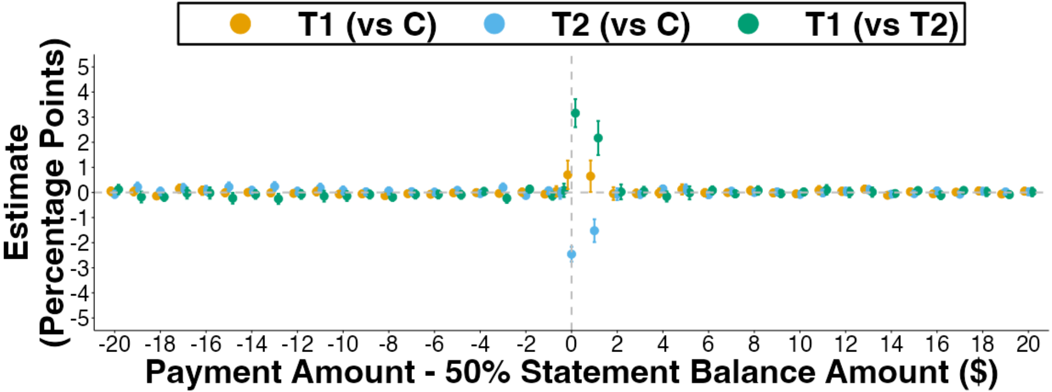




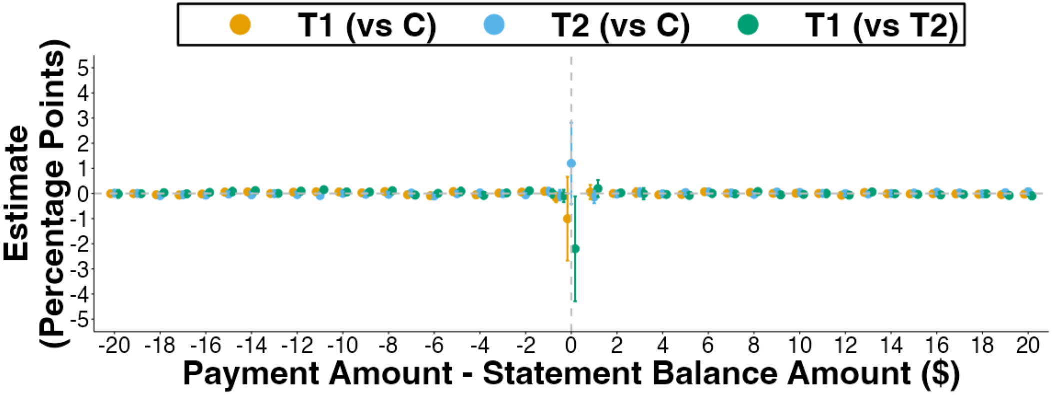
Effects On Heuristic Payment Amounts

Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)
Pay Multiples Of Minimum	-0.76***	-0.94***	0.18
	(0.25)	(0.23)	(0.29)
Pay Round Number	2.51***	3.61***	-1.10
	(0.62)	(0.62)	(0.80)
Time F.E.	X	X	X
Consumer Controls	X	X	X

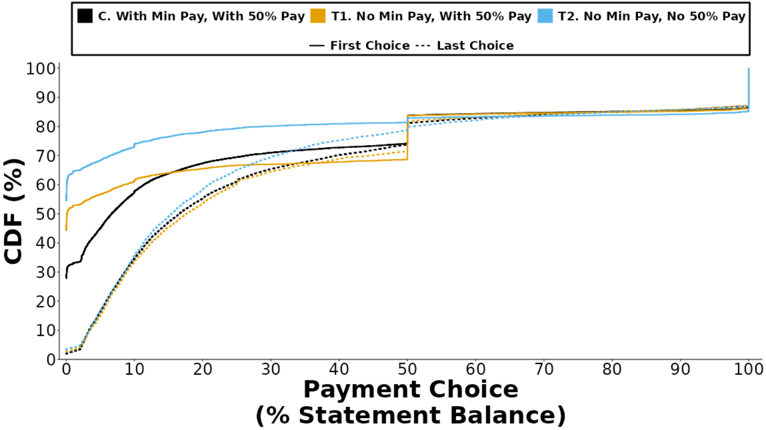
Effects On Payments Proximate To 50% Statement Balance Amount



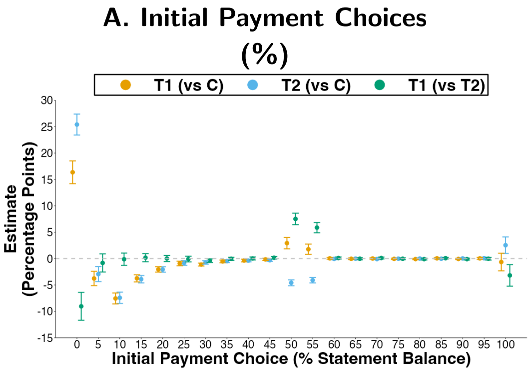
Effects On Payments Proximate To Statement Balance Amount

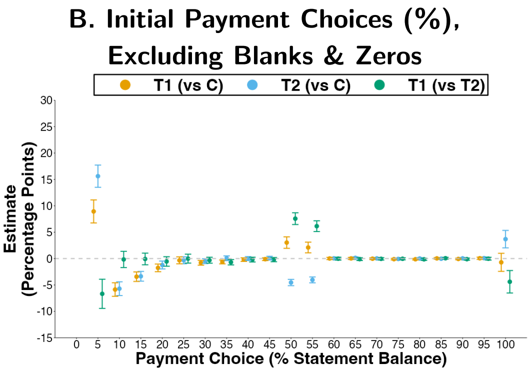


Initial vs. Last Payment Choices Within Payment Cycle



Effects On Initial Payment Choice





Initial Payment Choices

A. Initial Payment Choices

Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)
Initial Payment Choice (% Statement Balance)	-0.90	-4.33***	3.43***
	(0.87)	(0.81)	(1.07)
Initial Payment Choice Less Than Minimum Payment	19.44***	30.50***	-11.06***
	(1.16)	(1.02)	(1.38)
Number of Payment Choices	0.40***	0.85***	-0.46***
	(0.11)	(0.11)	(0.14)
Time F.E.	X	X	X
Consumer Controls	X	X	X

B. Initial Payment Choices, Excluding Blanks & Zeros

Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)
Initial Payment Choice (% Statement Balance)	-0.22	-1.82*	1.60
	(0.86)	(0.83)	(1.07)
Initial Payment Choice Less Than Minimum Payment	15.59***	23.57***	-7.98***
	(1.08)	(1.04)	(1.38)
Number of Payment Choices	0.40***	0.87***	-0.47***
	(0.11)	(0.11)	(0.14)
Time F.E.	X	X	X
Consumer Controls	X	X	X